

Implementing a Scalable, Secure Genomics Ingest and Processing Pipeline in a Trusted Research Environment

A governance-first infrastructure for discoverable, reusable dementia genomics

150,000+

participants

23

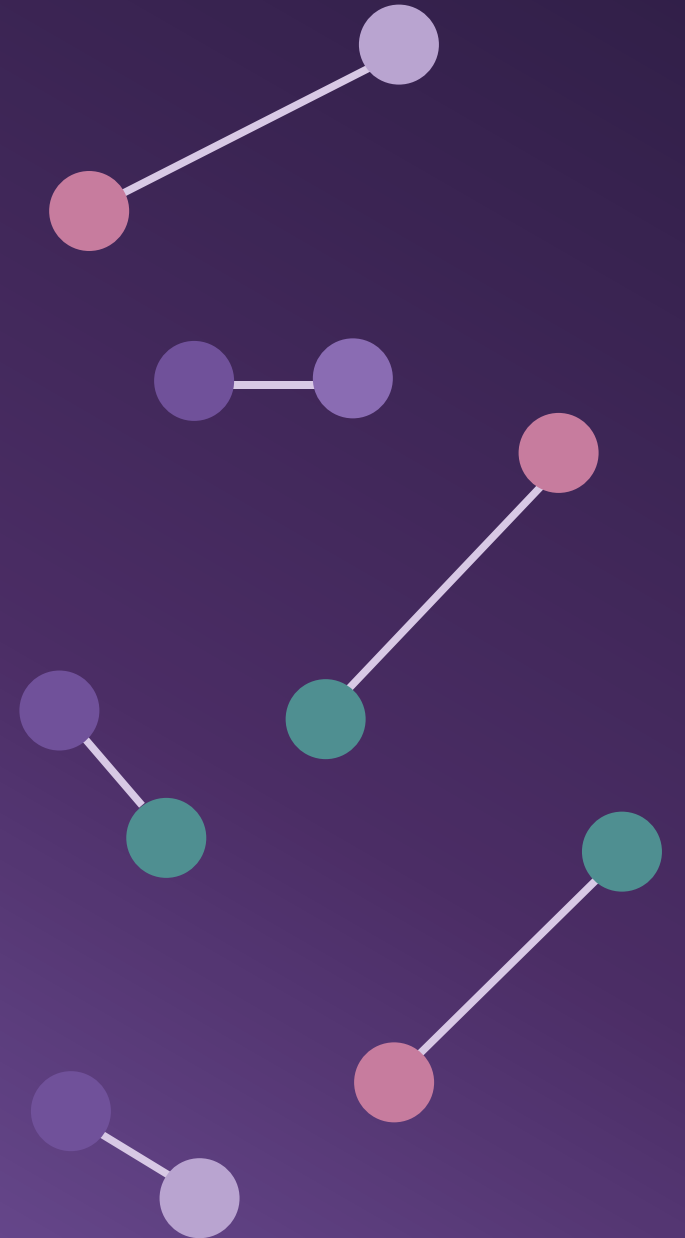
cohorts with genomics

20

directly held beyond gene status

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Why genomics needs more than secure storage

The data are scientifically powerful, but reuse depends on solving sensitivity, heterogeneity, computation and governance together.

1

Sensitive by nature

Individual-level genomic data can retain disclosure risk after conventional identifiers are removed.

2

Technically heterogeneous

Formats, assay types, genome builds and historical processing differ across cohorts.

3

Computationally intensive

Large files and tasks such as imputation require specialist software and scalable compute.

4

Governance-bound

Consent, access conditions and acceptable risk vary by cohort and scientific context.

The infrastructure contribution is the controlled path from heterogeneous files to research-ready data.

One repository, many genomics data types

The April 2026 resource snapshot combines hosted and brokered holdings across population, clinical and family cohorts.

55

cohorts in
the wider
DPUK
repository

23

cohorts
include
genomics

20

cohorts with
genomics
held directly

150,000+

participants represented

Primary molecular data

- **Array genotyping** PLINK / BGEN
- **Sequence variants** VCF / BCF
- **Methylation arrays** IDAT
- **RNA expression** RNA-seq profiles

Analysis outputs

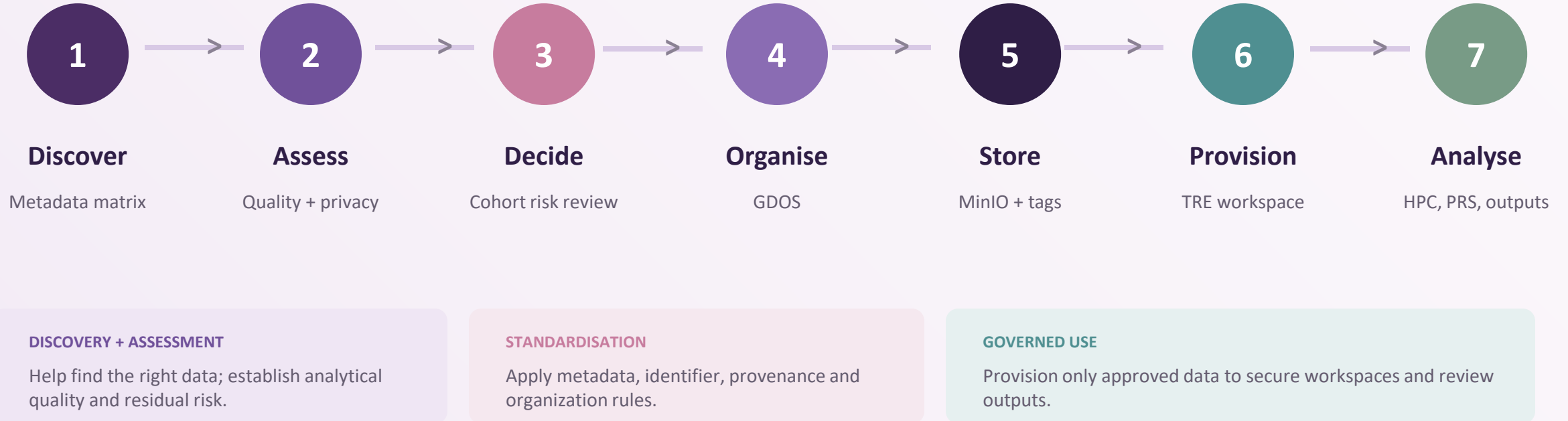
- **Genome-wide association** GWAS summary statistics
- **Population structure** Principal components (PCA)
- **Raw intensity data** Bespoke reprocessing

Derived and curated products

- **Polygenic risk scores** PRS
- **Gene-status variables** e.g. APOE
- **Linked context** Phenotype and imaging where available

A seven-stage path from discovery to approved analysis

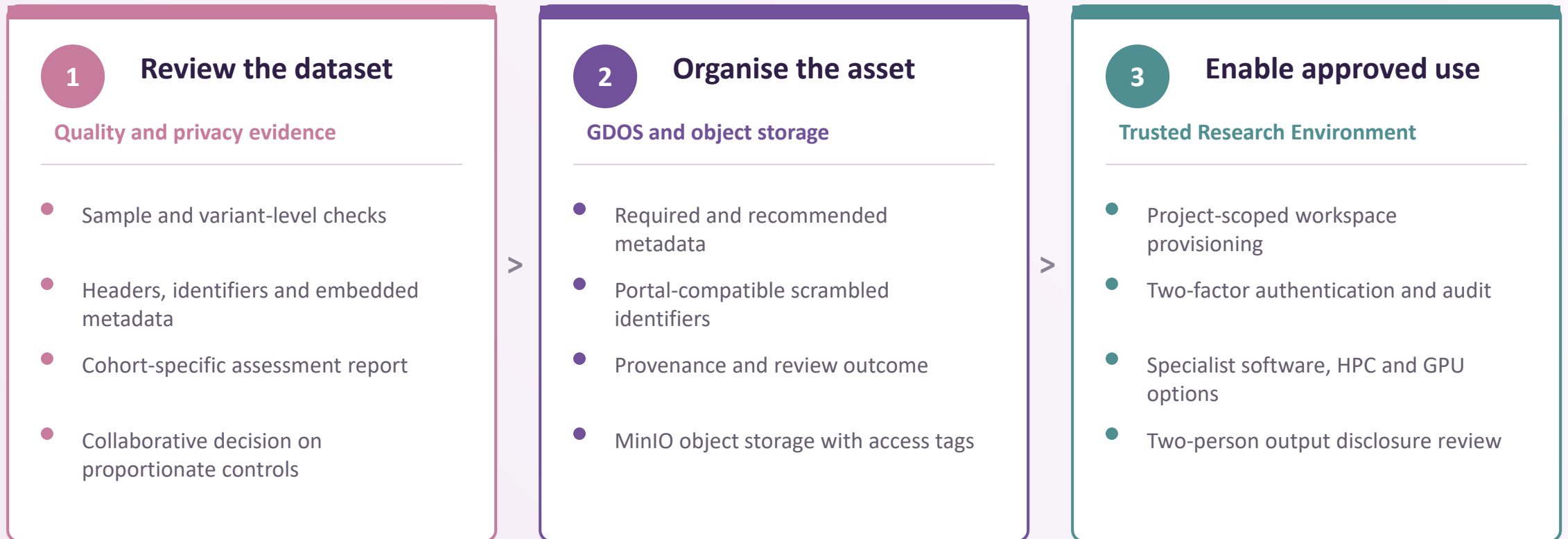
Each stage removes a different barrier: uncertainty, quality risk, disclosure risk, inconsistency, scale or access friction.



Data are visited inside the Trusted Research Environment; individual-level genomics files are not distributed to researchers.

Three operational layers make reuse possible

The architecture connects cohort-level decisions to organised storage and project-scoped analysis.



MinIO = scalable object storage | HPC = high-performance computing | TRE = Trusted Research Environment | GDOS = genomic data organization standard

GDOS is the contract around every dataset

The Genomic Data Organisation Standard covers description, linkage, traceability and location - not only file names.

DESCRIBE

Metadata profile

Required and recommended fields: data type, assay/platform, processing level, format and access state.

LINK

Identifier handling

Portal-compatible scrambled identifiers allow linkage without exposing original cohort identifiers.

TRACE

Provenance + quality

Source, processing method, review status and quality/privacy outcome travel with the dataset.

LOCATE

Hierarchy + naming

A predictable folder and filename pattern separates raw, quality-controlled and derived assets.

MANDATORY STRUCTURE

Directory > `/ {cohort} / omics / {modality} / {assay_type} / {processing_level} / {file_format} / <platform>`

File Name > `{cohort}_{modality}_{assay_type}_{processing_level}_{file_format}`

OPTIONAL SUFFIX TAGS

build

chromosome

platform / instrument

version / date

cohort-specific label

"Assay type" is used for the scientific measurement; "genotyping platform" records the specific array or instrument where known.

`/bio_uk/omics/dna/genotyping/raw/vcf/affymetrix_axiom`

Standards-informed, TRE-specific, implementation-ready

International standards provide reusable concepts; DPUK adds the governance and linkage context needed for a dementia Trusted Research Environment.

REFERENCE CONCEPTS

Minimum information, discovery, dataset description and provenance

GSC / MixS

GA4GH Data Connect

Bioschemas

RO-Crate

FAIR

DPUK REQUIREMENTS

Non-disclosive metadata that reflects governance and linked-data use

cohort-owner review

scrambled IDs

QC / privacy outcome

Five Safes

linked modalities

PLANNED METADATA LAYER

A validated, queryable profile connected to existing storage and provisioning

required / recommended fields

controlled vocabularies

Australian BioCommons-aligned minimum items

Gen3-compatible catalogue and API

MinIO remains the object-storage layer

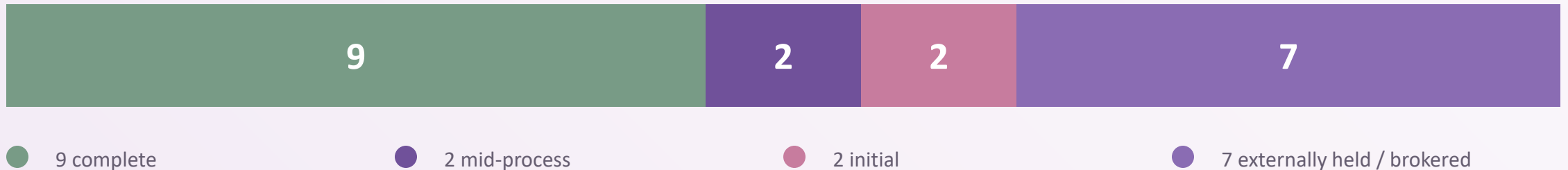
Planned work is labelled explicitly: the Gen3-compatible metadata layer and Australian BioCommons alignment are not presented as completed.

Quality and privacy review is a collaborative decision process

Technical checks produce evidence; cohort teams and DPUK then agree proportionate controls within the Five Safes model.



Current status across 20 cohorts with genomics beyond gene status



Five Safes is embedded across people, projects, settings, data and outputs - not added after technical processing.

From raw genotype files to a reusable risk-score product

A polygenic risk score (PRS) combines many small genetic effects into a relative score. It supports research comparison; it is not a clinical diagnosis.



Five cohorts processed

Cohort	Participants	Genotyping platform
Brains for Dementia Research	570	NeuroChip
Generation Scotland	20,032	Illumina ExomeArray
EPIC Norfolk	21,041	Axiom 2020
MRC National Survey of Health & Development	2,864	NeuroX2
Airwave 1 + 2	19,380	Affymetrix / Illumina

Disease and trait models

- Alzheimer's disease
- Parkinson's disease
- Frontotemporal dementia
- ALS / motor neurone disease
- Lewy body dementia
- Depression

Reference used: 1000 Genomes European population

Limitation: predominantly European ancestry reduces generalisability; ancestry must remain explicit in interpretation.

The contribution is governed reuse - not simply file storage

1

DISCOVERABLE

A cross-cohort view of data types, formats, platforms and processing state.

2

GOVERNED

Quality/privacy evidence, cohort-owner decisions and Five Safes access.

3

REUSABLE

GDOS metadata, identifiers, provenance and predictable organisation.

4

ANALYSABLE

Secure workspaces, specialist compute and shared products such as PRS.

A practical model for turning complex, high-volume linked data into a secure research resource.

Thank you

DPUK data access: data.dpuk.ukserp.ac.uk/Apply

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